FEUP

FACULTY OF ENGINEERING OF THE UNIVERSITY OF PORTO

Systems Engineering

Team Manual

SWAID - Plate Resonance (PR)

PLATE RESONANCE
TEAM

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1 Team Organization

1.1 Roles and Responsibilities

The team is organized with specific roles assigned to ensure efficient project execution. However, depending on project necessities and overall team performance throughout the semester, the team may deem it necessary to adjust these assigned roles or create new ones. The primary responsibilities are currently distributed as follows:

Role	Member	Primary Responsibilities
Team Leader	Miguel Amorim	Organize project development, coordinate with other SWAID teams, and ensure all deadlines and deliverables are met. Lead team meetings, oversee task assignments, and supervise overall performance based on set requirements. Act as the primary point of contact, report to the advising professor, manage budgeting, and collaborate with members on technical development.
Technical Leaders	João Oliveira (HW) Artur Kovbasa (SW)	Oversee repositories and technical documentation. Outline the project development plan—from concept to prototyping and task assignment—in collaboration with the Team Leader. Serve as technical points of contact for the Chladni Plate and Human Interface teams, ensuring technical requirements are met and inter-team challenges are resolved. Co-manage the development schedule.
Team Members	Afonso Cruz Alexandre Gorra Beatriz Martins Hugo Gomes João Pinto Lucas Aparicio	Contribute to technical development (Hardware/Software), documentation, and presentations while executing weekly assigned tasks. Proactively participate in project development through solution proposals and decision-making votes. Reach out and collaborate with internal members and other SWAID project colleagues to ensure overall project success.

Table 1: Team Roles and Responsibilities

1.2 Team Management

To ensure continuous progress and alignment, the team conducts two weekly meetings. The primary, in-person meeting takes place on Wednesdays to receive feedback from the advising professor, plan the upcoming week's tasks, and actively develop the project. Prior to this, a brief remote synchronization meeting is held to review the week's progress and properly prepare for the session with the professor.

For operational efficiency, the project is structured into two main sub-teams: FPGA/Hardware Development and Software Development. Each sub-team is guided by a technical leader who, alongside the general team leader, manages the project's technical execution. The hardware team focuses on FPGA programming and acts as the primary liaison with the Chladni Plate (CP) team. Conversely, the software team is dedicated to developing the software abstraction layer and collaborating directly with the Human Interface (HI) team.

Despite this targeted division, the team operates cohesively as a single unit. Core responsibilities such as documentation, system integration, and presentation preparation are shared across all members. Every member is expected to proactively contribute ideas, support their peers, and participate in critical decision-making processes throughout the project lifecycle.

1.3 Conflict Management

For the project to proceed as smoothly as possible, it is fundamental that all team members assume a serious commitment to the work and the team. Each member should strive to fulfill the assigned objectives and, whenever difficulties arise, act with transparency, clearly explaining the obstacles encountered.

It is also expected that there is a willingness to collaborate and support other colleagues whenever necessary, promoting a spirit of mutual help. The work should be developed with dedication, responsibility, and attention to detail, seeking to achieve results with the highest possible rigor and quality.

In case of technical, organizational, or strategic divergences that cannot be resolved through normal discussion, an internal vote will be held, with the final decision taken by a simple majority of the votes of the members present. In the event of a tied vote, the Team Leader is granted the deciding vote (vote of quality) to break the tie and finalize the decision.

2 Infrastructure and Collaborative Tools

To ensure seamless cooperation and traceability across the **Plate Resonance (PR)** group, our infrastructure is divided into three functional pillars: project management, version control, and standardized documentation.

2.1 Project Management

We utilize a suite of tools to coordinate high-level planning with daily team operations:

- **Github (Planning & Development):** The primary platform software development and project management, making use of the integrated Projects tool that includes Gantt diagram functionalities, Kanban boards, task assignment, issue management, and direct task to commit traceability.
- **Discord (Communication Hub):** Used for synchronous sprints and collaborative debugging. Integrated with GitHub for real-time updates.
- **WhatsApp (Instant Messaging):** Used for immediate announcements, quick technical queries, and internal team discussions.
- **When2Meet (Scheduling):** The standard tool for finding common availability for ad-hoc sync meetings.
- **Google Forms (Evaluation):** The official platform for conducting periodic peer assessments and gathering team feedback.

2.2 Workflow and Version Control Rules

To maintain a clean and traceable codebase and document history, all contributions must follow a centralized workflow.

2.2.1 Task Lifecycle

Project work is driven by the Github Projects timeline. Members may select available tasks or be appointed to specific ones by the team or technical leaders. Once assigned, these high-level tasks `[PR#ID]` are broken down into granular GitHub Issues (e.g., `GH#45`) to track technical

progress and specific bugs. Every member is free to choose their preferred development environment for code and text editing.

2.2.2 Contribution Rules

The repository is protected through strict Git protocols:

- **Pull Requests (PRs):** No code or text is merged directly. All work must be submitted via PRs, which are reviewed and merged exclusively by Technical Leaders to resolve potential conflicts.
- **Commit Naming Convention:** Commits must follow the format: `[PR#ID] [GH#ID] Brief, imperative description`.
- **Traceability:** Every PR description must link to the relevant Issue and Project Requirement.

2.3 Documentation

Standardization is critical to ensure that our technical reports are presented as a unified subsystem.

2.3.1 Standardized Templates

All document templates include a standardized cover (FEUP symbol, SWAID project name, PR identification), a complete team roster with student numbers, and automated navigation elements (ToC and page numbering).

It is highly recommended that members consult the `docs/templates/` directory and read the detailed documentation guidelines provided there. Every document must include the designated standard preamble found in the templates, which defines the essential packages and formatting required for consistency. While additional packages may be added depending on specific document requirements, members should consult the documentation guidelines for further details on compatibility and standards.

2.3.2 Modular Architecture

To allow multiple members to work on the same report simultaneously without Git conflicts, we use a Modular LaTeX Structure:

- Content is organized into separate `.tex` files within the `/src` folder.
- **Naming Convention:** File names must start with a numerical prefix (e.g., `01_intro.tex`) corresponding to their order of inclusion in the `main.tex` file. This ensures the repository file list matches the document structure.

2.3.3 Compilation and Environment

While the choice of editor is personal, the following technical standards are mandatory for the final output:

- **XeLaTeX:** All official documentation must be locally compiled using the XeLaTeX engine to ensure correct rendering of the Noto font family and modern packages.

- **Recommended Setup:** For the best experience with our modular structure, we recommend VS Code with the *LaTeX Workshop* extension for its robust synchronization and linting capabilities.

3 Evaluation Criteria and Process

As per course requirements, each team member will be periodically evaluated by their peers. At the end of the project, an individual final grade will be calculated based on the combination of peer assessments and official meeting attendance. This grade reflects each member's technical contribution, reliability, and commitment to the project's success.

3.1 Evaluation Process

The evaluation process consists of a confidential peer assessment conducted every two weeks (six cycles in total). Each team member evaluates all other members according to three qualitative criteria, while attendance is tracked quantitatively.

3.1.1 Peer Assessment Score

Each qualitative criterion (C_n) is rated on a scale from 1 (low) to 5 (excellent). For each evaluation cycle, a Member Score (MS) is calculated as the sum of these three criteria:

$$MS = C_1 + C_2 + C_3$$

The maximum score per cycle from a single peer is 15. At the end of the semester, the average of all cycles is calculated:

$$\text{Peer Average} = \frac{1}{6}\sum_{i=1}^{6}\left(\frac{1}{10}\sum_{j=1}^{10} MS_j\right)_i$$

3.1.2 Attendance Score (A)

Attendance is strictly recorded for the 13 scheduled weekly Wednesday meetings with the supervisor. For each of these meetings, a member receives 1 point for being present and 0 points for an absence. The total is then scaled to a 5-point range using the following formula:

$$A = \frac{5}{13}\sum_{k=1}^{13} \text{Presence}_k$$

While only the Wednesday meetings contribute to the quantitative Attendance Score, attendance for all other internal scheduled meetings (remote or co-working) is recorded. These records serve as a reference to assist team members in accurately evaluating their peers in the qualitative categories (Commitment, Proactivity, and Teamwork).

3.1.3 Final Grade Calculation

The final individual evaluation grade is the sum of the Peer Average and the Attendance Score, resulting in a total possible score of 20:

$$\text{Final Grade} = \text{Peer Average} + A$$

To ensure fairness, any qualitative rating of 3 or less must be justified. Significant discrepancies or persistent issues identified during assessment cycles will be addressed in internal meetings to ensure conflicts are managed constructively and do not unfairly bias a member's score.

3.1.4 Final Normalization

For the final report, internal scores are converted into a relative evaluation table. This table must follow three strict rules:

- **Mean = 0:** The sum of all individual evaluations in the team must be zero.
- **Range:** All values must fall within the interval $[-1.5, 1.5]$.
- **Spread Requirement:** At least one member must receive a $+1$ and at least one must receive a -1.

The conversion from the Internal Score (S_i) to the Final Evaluation Factor (V_i) is calculated by determining the deviation from the team mean ($\bar{S}$) and scaling it:

$$V_i = \text{Scale} \times (S_i - \bar{S})$$

The *Scale* factor is adjusted at the end of the semester to ensure the resulting values satisfy the ± 1 requirement and remain within the $[-1.5, 1.5]$ bounds.

3.2 Evaluation Criteria

The qualitative criteria used for peer assessment are:

- **Commitment and Responsibility (C_1):** Reliability in meeting internal deadlines, quality of assigned deliverables, and the ability to fulfill responsibilities without constant supervision.
- **Proactivity and Contribution (C_2):** Initiative in proposing solutions, autonomy in technical development, and active participation in the decision-making process.
- **Teamwork and Communication (C_3):** Clarity in reporting progress, availability to support colleagues, and effective use of collaborative tools (GitHub, Discord, WhatsApp).
- **Official Attendance (Quantitative):** Verified presence at the 13 mandatory Wednesday sessions with the supervisor, demonstrating professional consistency and alignment with the project's formal requirements.